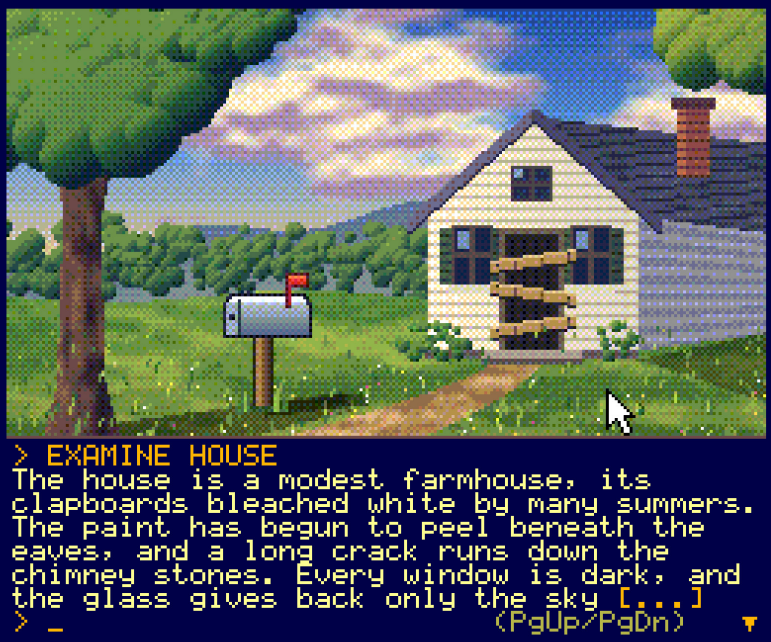
# UI specs

Illustrated interactive fiction · Zephyr-80 / KonaOS · V9958 G7

Coordinates are native pixels, with the origin at the top left. Each rule is marked **Brief** (from the design brief), **Decided** (settled in review) or **Proposed** (a default, still open). Nothing here has been run on the machine yet.




## Screen

| Element | Spec | Status |
|---|---|---|
| Mode | G7, 256×212, 8 bits per pixel, fixed `GGGRRRBB` colours | Brief |
| Scene | 256×144 at `y 0–143`, 68% of the screen. Replaces the brief's 75% (256×159) to give the text more room. | Decided |
| Divider | `y 144`, 1 line | Proposed |
| Text panel | `y 145–211`, dark background | Brief |
| Narrative rows | 7 rows, tops at `y 147, 155, 163, 171, 179, 187, 195` | Decided |
| Prompt row | Top at `y 203`, always the last row | Brief |
| Text columns | 42, from `x 2` to `x 253` | Decided |
| Fit | 1 divider line + 2 px top margin + 8 rows × 8 px + 1 px bottom margin = 68 lines, the whole panel with none left over. | Measured from the mockup |

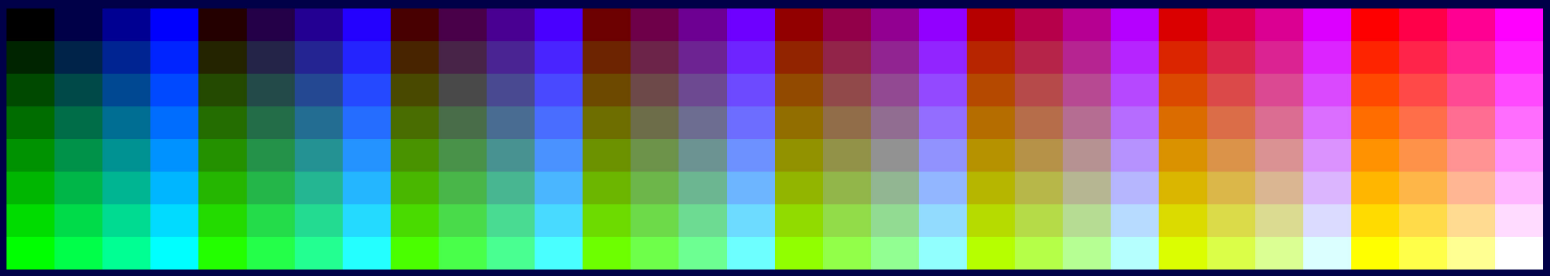
## Font and colour

6×8 cell, chosen over the 4×6 test for readability. Lowercase `g j p q y` use the bottom row as a descender. The arrows are `0x1E` and `0x1F` in CP850; the mockups draw stand-ins, so the real glyphs come from your font.

| Use | G7 byte | Mockup RGB | Status |
|---|---|---|---|
| Panel background, highlighted menu text | `0x01` | `#000049` | Proposed |
| Divider | `0x4D` | `#6D4949` | Proposed |
| Narrative text, menu text and frame | `0xFE` | `#FFFF92` | Proposed |
| Prompt, echoed command, `[...]`, arrows, menu highlight | `0xBC` | `#FFB600` | Decided for `[...]`, otherwise proposed |
| `(PgUp/PgDn)` hint | `0xB5` | `#B6B649` | Proposed |
| Menu background | `0x25` | `#242449` | Proposed |
| Menu shadow | `0x00` | `#000000` | Proposed |

## Colour palette

**Bitmap: fixed.** In G7 each pixel byte is its own colour, `GGGRRRBB`, so there is no lookup and the palette registers don't apply. All 256 colours are available everywhere, with no per-scene palette to manage. Below, green rises by row, and each column block of 4 is one red level with blue rising left to right.



**Sprites: also fixed, but a different set.** In G7, sprite colours come from the 16 codes below and are not related to the palette registers. Only these colours are available to the cursor.

| Code | Colour | G | R | B | Mockup RGB |
|---|---|---|---|---|---|
| `0` | | 0 | 0 | 0 | `#000000` |
| 1 | | 0 | 0 | 2 | `#000049` |
| 2 | | 0 | 3 | 0 | `#6D0000` |
| 3 | | 0 | 3 | 2 | `#6D0049` |
| 4 | | 3 | 0 | 0 | `#006D00` |
| 5 | | 3 | 0 | 2 | `#006D49` |
| 6 | | 3 | 3 | 0 | `#6D6D00` |
| 7 | | 3 | 3 | 2 | `#6D6D49` |
| 8 | | 4 | 7 | 2 | `#FF9249` |
| 9 | | 0 | 0 | 7 | `#0000FF` |
| 10 | | 0 | 7 | 0 | `#FF0000` |
| 11 | | 0 | 7 | 7 | `#FF00FF` |
| 12 | | 7 | 0 | 0 | `#00FF00` |

| | | | | | |
|---|---|---|---|---|---|
| 13 | | 7 | 0 | 7 | #00FFFF |
| 14 | | 7 | 7 | 0 | #FFFF00 |
| 15 | | 7 | 7 | 7 | #FFFFFF |

Code 0 is transparent unless TP (R#8 bit 5) is set. RGB values throughout this page are the mockup's linear estimates (bitmap blue uses levels 0, 2, 4, 7 of 7), not measurements of the LunchCrema output.

## Long text

| Rule | Spec | Status |
|---|---|---|
| Paging | A response longer than the panel stops at its first page. There is no `[MORE]` line and no scroll bar. | Decided |
| Continuation marker | Amber `[...]` after the last word of a continued page, separated by one space. Not shown on the last page. Large enough to notice on the LCD. | Decided |
| Wrapping | 42 columns, except the last row of a continued page, which wraps at 36 so the marker never covers text. | Proposed |
| Page overlap | None. Each page starts on new text. | Proposed |
| Command echo | The echoed command uses the first narrative row of the response, in amber. | Proposed |
| Keys | PgUp and PgDn move one page. | Decided |
| Hint | `(PgUp/PgDn)` at column 27 of the prompt row, shown while the response has more than one page. | Decided |
| Arrows | ▲ at column 39, ▼ at column 41. Fixed positions, blank when that direction is unavailable. Large enough to notice on the LCD, and clickable. | Decided |
| Arrow hit areas | ▲ `x 232–243`, ▼ `x 244–255`, both `y 202–211`. | Proposed |
| Typing while paging | Any character jumps to the last page and goes into the prompt. | Proposed |
| Long commands | Up to 24 cells of input and cursor with the hint showing. Past that the hint hides (36 cells), then the arrows hide (40 cells). | Proposed |
| After a long response | The hint stays until the next command. If the next response fits, the hint goes and only ▲ remains, for reaching earlier text. | Proposed |
| History | Word-wrapped text in a RAM ring buffer. The panel is redrawn from it when paging. | Proposed |

## Context menu

| Rule | Spec | Status |
|---|---|---|
| Trigger | Clicking an obvious object. Hidden objects and secret interactions never get a menu. | Brief |
| Items | General verbs only, such as `EXAMINE`, `TAKE`, `OPEN`, `PUSH`, `PULL`, in uppercase. | Brief |
| Result | Choosing an item submits the matching command, such as `OPEN MAILBOX`, and the response appears in the panel. | Brief |
| Size | 50 px wide, 5 px of padding in total plus 9 px per item, so 32 px for three items. 7-letter labels fit. | Proposed |
| Style | 1 px frame, 2 px offset shadow, 5 px tail toward the object, full-width amber highlight under the cursor. | Proposed |
| Placement | Beside the object, choosing the side that hides the least important art. In Mockup B that is above, at `x 70, y 53`. | Proposed |

| | | |
|---|---|---|
| Closing | A click elsewhere or Esc closes the menu without submitting. Typing closes it and goes to the prompt. | Proposed |
| Restoring the scene | Copy the covered area back from a clean copy of the scene on VRAM page 1. | Proposed |

## Cursor and scenes

| Rule | Spec | Status |
|---|---|---|
| Cursor | 12×16 arrow, drawn with two overlaid 16×16 hardware sprites so the bitmap is never touched. Fill is sprite code 15 (white). A true black outline is code 0, which needs TP=1; code 1 (dark blue) avoids setting TP. | Proposed |
| Hotspots | Never drawn. No hover highlight or floating indicators. | Brief |
| Scenes | Static art, with no sprites, scrolling or animation. Art may change when the world state changes, such as an opened mailbox. | Brief |
| Story-state art | Each story state has a single image asset, chosen by the game when it detects that state. Visual cues such as the mailbox flag are a game-level decision, not a UI rule: if a cue matters to the story, it becomes a state with its own image. | Decided |
| Permanent UI | No inventory sidebar, toolbar or decorative frame. | Brief |
| Memory per scene | 36,864 bytes uncompressed (256 × 144). | Measured from the mockup |

## Still open

Panel redraw and menu restore times have not been measured on the machine yet. Scene compression has not been decided.